

# Farahnaz Hosseini

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## Education

- **Ph.D. in Computer Science**

University of North Texas (UNT)

January 2022- May 2026

- Dissertation Title: " **Graph Neural Networks for Neighborhood Socioeconomic Modeling from High-Dimensional Census Data**"

- Advisor: Dr. Ting Xiao

- **Master of Science in Computer Engineering**

University of Kashan

September 2012 - June 2014

- Thesis Title: " **Brain Tumor Detection and Recognition in MRI Images**"

- Advisor: Dr. Hossein Ebrahimpour Komleh

- **Bachelor of Science in Computer Engineering**

Kerman Higher Institute

September 2008 - June 2010

- **Associate Degree in the field of Computer Hardware**

Kharazmi Applied-scientific Higher Education Institute-Tiran and Karvan Branch, Tiran September 2004 - September 2008

## Research Interest

- Artificial Intelligence
  - Machine Learning / Deep Learning
  - Graph Neural Networks
  - Vector Embeddings for AI Applications
  - Multimodal Large Language Models
  - Clinical and Biomedical Applications of AI
  - Big Data/ Data Science
  - High Performance Computing
  - Cybersecurity
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## Publications

(20+ publications, 53+ citations, 5 h-index)

1. Hosseini, F., and T. Xiao." **Graph Neural Networks and Geo-Embedded Census Data for Socioeconomic Prediction: A Comprehensive Review**", *Academia AI and Applications Journal* 2026, USA.
2. Hosseini, F., and T. Xiao. "**Robust Socioeconomic Community Discovery from Census-Derived Indicators via KNN Graphs and Louvain Stability-Conductance Analysis**", *IEEE SoutheastCon* 2026. Huntsville. Alabama, USA.

3. **Hosseini, F.**, and T. Xiao. 2025. "Discovering Socioeconomic Structure with Unsupervised Graph Neural Embeddings on Census Data." *12th Annual Conference on Computational Science & Computational Intelligence (CSCI 2025) Las Vegas, USA*.
4. **Hosseini, F.**, and T. Xiao. 2025. "Graph-Augmented Machine Learning for Socioeconomic Prediction Using Census Data." *12th Annual Conference on Computational Science & Computational Intelligence (CSCI 2025) Las Vegas, USA*.
5. **Hosseini, F.**, and T. Xiao. 2025. "Using Dynamic Graph Neural Networks for Temporal Analysis in Financial Markets." *4th International Conference on Computing and Machine Intelligence (CMI 2025), Michigan, USA*.
6. Nickfarjam, A. M., M. Dalilian, S. A. Zargari, M. Farzandipour, **F. Hosseini**, and T. Xiao. 2025. "Comparison of Deep Learning Methods for Age Determination in Forensic Clients Aged 5–18 Years." *4th International Conference on Computing and Machine Intelligence (CMI 2025) Michigan, USA*.
7. Nickfarjam AM, **Hosseini F**, Kia ZS, Shokrizadeh Arani L, Xiao T. "White Blood Cell Detection and Counting Using PSO". *Stud Health Technol Inform*. 2025 Jun 26;328:332-336. doi: 10.3233/SHTI250731. PMID: 40588938.
8. Nickfarjam, A. M., **F. Hosseini**, and T. Xiao. 2025. "White Blood Cell Detection and Counting Using Particle Swarm Optimization." *23rd International Conference on Informatics, Management, and Technology in Healthcare*.
9. **Hosseini, F.**, and T. Xiao. 2025. "Developing K-Core and K-Truss Algorithms in Large Dynamic Graphs." In *Proceedings of the SIAM Conference on Computational Science and Engineering (CSE25)*, Fort Worth, TX.
10. Khanda, A., S. M. Shovan, A. Pandey, **F. Hosseini**, and S. K. Das. 2024. "Efficient Approaches to Analyzing Large Dynamic Networks." In *Proceedings of SC24: The International Conference for High Performance Computing, Networking, Storage, and Analysis*, Atlanta, GA.
11. Biglari, S., **F. Hosseini**, A. Upadhyay, and H. Zhao. 2024. "Survey of Network-on-Chip (NoC) for Heterogeneous Multicore Systems." In *Proceedings of the 17th IEEE International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoc 2024)*.
12. Bevara, R. V. K., I. Wagenvoort, **F. Hosseini**, H. Sharma, V. Nunna, and T. Xiao. 2024. "Census2Vec: Enhancing Socioeconomic Predictive Models with Geo-Embedded Data." In *Proceedings of the Intelligent Systems Conference (IntelliSys 2024)*, 626–640. Springer Nature.

13. Rafati, M., N. Kalantari, J. Azadbakht, A. M. Nickfarjam, and **F. Hosseini**. 2024. "Performance Evaluation of Fuzzy Genetic, Fuzzy Particle Swarm and Similar Insects' Optimization Algorithms on Denoising Problem Based on a Novel Combined Filter for Digital X-ray Images." *Multimedia Tools and Applications* 83 (6): 15483–15531.
14. Zarei, M. A., **F. Hosseini**, A. M. Nickfarjam, and T. Xiao. 2023. "Predicting User Interest Using Hierarchical-Based Clustering for Recommender Systems." In *Proceedings of the International Conference of Computer Science and Information Technology (ICCSIT 2023)*.
15. Bevara, R. V. K., T. Xiao, and **F. Hosseini**. 2023. "Bias Analysis in Language Models Using an Association Test and Prompt Engineering." In *Proceedings of the 2023 IEEE International Conference on Software Quality, Reliability & Security (QRS 2023)*.
16. Rafati, M., F. Farnia, E. Romoozi, A. M. Nickfarjam, and **F. Hosseini**. 2021. "Measuring Panoramic Radiomorphometric Indices for Mandible Bone Using Active Shape Model and BIC-SVM." *International Journal of Imaging Systems and Technology* 31 (3): 1711–1730.
17. Nickfarjam, A. M., H. Ebrahimpour-Komleh, and **F. Hosseini**. 2015. "Multi-level Image Thresholding Based on Local Variance and Particle Swarm Optimization." In *Proceedings of the International Congress on Technology, Communication and Knowledge (ICTCK 2015)*.
18. **Hosseini, F.**, H. Ebrahimpour-Komleh, and M. KhodamHazrati. 2015. "Quantitative Evaluation of Skull Stripping Techniques on Magnetic Resonance Images." In *Proceedings of the World Congress on Electrical Engineering and Computer Systems and Science (EECSS 2015)*.
19. Askari, M., H. Ebrahimpour, A. A. Bidgoli, and **F. Hosseini**. 2013. "Parallel GPU Implementation of Hough Transform for Circles." *International Journal of Computer Science Issues* 10 (2): 215–221.

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## Professional Experience

### Graduate Research Assistant

**University of North Texas, Denton, Texas, USA | January 2022 – May 2025**

- Conducted NSF-funded research on scalable k-core and k-truss decomposition algorithms for large dynamic graphs using high-performance computing (HPC) methods.
- Developed graph-based and deep learning methods for socioeconomic predictive modeling using graph neural networks (GNNs) and multi-source data.
- Designed and evaluated machine learning and graph learning pipelines for large-scale data analysis and predictive modeling.

### Teaching Experience

#### Computer Engineering Instructor and Lecturer

**Payame Noor University | January 2015 – May 2015**

- Taught courses in Artificial Intelligence and Research Methods.

- Developed course materials, assessments, and instructional content.
- Supervised undergraduate student projects and provided academic guidance.

## **Instructor**

### ***Technical and Professional Organization* | August 2007 – 2020**

- Taught courses in programming languages, web development, software engineering, database systems, operating systems, networking and security, and software tools.
- Delivered lectures, prepared instructional materials, and evaluated student performance through exams and projects.
- Provided detailed feedback and supported student skill development across technical and applied computing topics.

## **Graduate Teaching Assistant**

### ***University of North Texas, Denton, Texas, USA* | May 2025 – May 2026**

- Supported instruction in Social Issues in Computing and Data Visualization courses.
  - Assisted students with hands-on projects using Python, Matplotlib, Plotly, Tableau, Power BI, and D3.js, and contributed to grading, feedback, and tutorial support.
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## **Advising and Mentoring Experience**

- **Academic Mentor, UR2PhD Program, Computing Research Association**  
Mentored undergraduate students in research methods, graduate school preparation, and technical topics including parallel programming, CUDA, OpenMP, and graph algorithms.
  - **Graduate Advisor, AI Lab Summer Program**  
Advised graduate students on AI and data science summer research projects, including project planning, technical development, and academic guidance.
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## **Proposal and Grant Writing Support**

- Contributed to the development and writing of research proposals and supporting documents for grant applications in AI, machine learning, and data science.
  - Assisted with technical writing, literature synthesis, and proposal preparation for funded and seed-grant research projects.
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## **Awards and Honors**

### **Awards, Honors, Grants, and Academic Service**

- Toulouse Graduate School Grant, University of North Texas, 2025.
- Department of Computer Science and Engineering Grant, University of North Texas, 2025.
- CSCE Grant, University of North Texas, 2025.
- ACM SIGHPC Grant Recipient, Supercomputing 2024 (SC24), Atlanta, Georgia, USA, 2024.
- Student Volunteer, Supercomputing 2023 (SC23), Denver, Colorado, USA, 2023.
- Student Volunteer, Supercomputing 2022 (SC22), Dallas, Texas, USA, 2022.
- Undergraduate Mentor, Computing Research Association (CRA) UR2PhD Program, 2023–2024.
  - Mentored two undergraduate students on research projects involving algorithms for large-scale dynamic graphs.
- Top Graduating Student, Department of Computer Engineering, University of Kashan.
- Thesis recognized in written media as a successful joint collaboration with the University of Florida in brain tumor detection and recognition using MRI.
- Achieved a top 25% worldwide ranking in the KDD Cup 2013 “Author Disambiguation Challenge” (Track 2) 2013.
- Certificate of Achievement, ACM International Collegiate Programming Contest (ICPC), Asia Tehran Regional, 2006.
- Certificate of Achievement, 6th Place, ACM International Collegiate Programming Contest, Asia Tehran Regional, 2013.

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## Oral Presentations and Posters

1. **Hosseini, F. (2025).** *Developing k-core and k-truss algorithms in large dynamic graphs*. Oral presentation, SIAM Conference on Computational Science and Engineering (CSE25), Fort Worth, TX.
2. Pandey, A., Khanda, S., Shovan, F. M., **Hosseini, F.**, Das, S., Norris, B., & Bhowmick, S. (2024). *Efficient approaches to analyzing large dynamic networks*. Poster presentation, The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC24).
3. Pandey, A., Khanda, S., Srinivasan, S., Srinivasan, S., Shovan, F. M., **Hosseini, F.**, Das, S., Norris, B., & Bhowmick, S. (2023). *Scalable algorithms for analyzing large dynamic networks using CANDY*. Poster presentation, The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC23).
4. **Hosseini, F.**, Ebrahimpourkomleh, H., & Nikfarjam, A. M. (2014). *Image segmentation using pixel mapping to a new space*. National Conference on Science and Computer Engineering, Najaf Abad. Paper No. 1245.
5. **Hosseini, F.**, Ebrahimpourkomleh, H., & Nikfarjam, A. M. (2014). *Large intestine cancer data evaluation and classification*. 2nd Conference on Computer and Information Technology, Tabriz.
6. **Hosseini, F.**, Rezaee, Z., & AzimiKashani, A. (2014). *Real-time data processing in wireless sensor networks*. National Conference on Computer Engineering and Information Technology, Shoushtar. Paper No. 54327.322.
7. **Hosseini, F.**, Ebrahimpourkomleh, H., & Askari, M. (2014, June). *A novel method with high accuracy for signature detection using PCA algorithm*. 4th International Conference on Information Technology Management, Communication and Computer (MITC2014), Tehran. Paper No. 93.104.214.
8. **Hosseini, F.**, Ebrahimpourkomleh, H., & Askari, S. (2014, June). *A novel method for iris recognition using BP neural network and parallel computing with the aid of GPU*. 4th International Conference on Information Technology Management, Communication and Computer (MITC2014), Tehran. Paper No. 93.104.214.
9. Nikfarjam, A. M., Ebrahimpourkomleh, H., & **Hosseini, F.** (2014, November). *Multi-level image thresholding based on local variance and particle swarm optimization*. 2nd International Conference on Intelligent Information Systems and Complex Systems, Mashhad. Paper No. 52.

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## Professional Service

- Member, **Association for Computing Machinery (ACM)**, 2022–Present
- Member, **Institute of Electrical and Electronics Engineers (IEEE)**, 2021–Present
- Member, **Computing Research Association (CRA)** 2023-Present
- **Member, AnitaB.org**, 2022–Present
- **Member, Women in HPC (WHPC)**, 2024–Present

## University Service

- Graduate Advisor, **UNT AI Summer Research Program**, 2023–Present

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## Book Chapters

Translated 1 chapter of the book “Probabilistic Robotics”, Sebastian.Thrun, ISBN-13: 978- 0262201629, 2013

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## Conference Reviewer:

**QRS 2026: The 26th International Conference on Software Quality, Reliability, and Security.** Florence Italy. July 22-25, 2026.

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## **Certificates**

- **Google Cybersecurity Professional Certificate by Google**
- **Google AI Essential Professional Certificate by Google**
- **Duke University Introduction to LLMOps with Azure**
- **Duke University Open Source LLMOps Solutions**
- **Duke University Developing on AWS for AI**
- **Duke University Large Language Model Operations (LLMOps) Specialization**
- **Duke University GenAI and LLMs on AWS**
- **Duke University Advanced Data Engineering**
- **Duke University Databricks to Local LLMs**
- **Certificate of Achievement, ACM International Collegiate Programming Contest, Asia Tehran Regional 2009**
- **Certificate of Achievement, ACM International Collegiate Programming Contest, Asia Tehran Regional 2013**
- **Workshop at the MITC 2014 International Conference on Information Technology Management, Communication and Computer 2014**